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ОТЧЕТ
к лабораторной работе №2
на тему:
«РАБОТА С DOCKER»
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1. Подготовка рабочего окружения

1. а. Установка Docker

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker --version
Docker version 29.1.4, build 0e6fee6
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker compose version
Docker Compose version v5.0.1
```

1. б. Выполнение базовых настроек

Докер установлен, но не работает. Перед **первым запуском команды Docker** необходимо запустить демон docker:

```
sudo systemctl start docker.service
```

Автоматический запуск Docker даемон при загрузке системы:

```
sudo systemctl enable docker.service
```

Однако по умолчанию для выполнения команд **docker** необходимо использовать **sudo**. Чтобы избежать использования **sudo** при выполнении каждой команды **docker**, можно добавить себя (или любого другого пользователя) в группу **docker** следующим образом:

```
sudo usermod -aG docker $USER
```

Чтобы вышеуказанное изменение вступило в силу, необходимо выйти из системы (или закрыть терминал) и снова войти в нее. Если вы не хотите этого делать, воспользуйтесь следующей командой:

```
newgrp docker
```

Проверка работоспособности Docker:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.
```

2. Изучение простейших консольных команд и возможностей Docker Desktop:

```
Common Commands:
run      Create and run a new container from an image
exec     Execute a command in a running container
ps       List containers
build    Build an image from a Dockerfile
bake     Build from a file
pull     Download an image from a registry
push     Upload an image to a registry
images   List images
login    Authenticate to a registry
logout   Log out from a registry
search   Search Docker Hub for images
version  Show the Docker version information
info     Display system-wide information

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker info
Client: Docker Engine - Community
Version: 29.1.4
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.30.1
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v5.0.1
    Path: /usr/libexec/docker/cli-plugins/docker-compose

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED
STATUS        PORTS     NAMES
291f0b9314f0   docker/getting-started "/docker-entrypoint..." 19 seconds ago
Up 19 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp nostalgic_wilson

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED
STATUS        PORTS     NAMES
5e2e2a614ea2   docker/getting-started "/docker-entrypoint..." 20 seconds ago
Created                                              wonderful_hoover
291f0b9314f0   docker/getting-started "/docker-entrypoint..." 27 seconds ago
Up 26 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp nostalgic_wilson
5a81481f7465   hello-world "/hello"                About an hour ago
Exited (0) About an hour ago                       stupid_davinci
401dab929337   hello-world "/hello"                About an hour ago
Exited (0) About an hour ago                       crazy_torvalds

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker ps -q
291f0b9314f0

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker ps -qa
5e2e2a614ea2
291f0b9314f0
5a81481f7465
401dab929337

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker images
IMAGE                                ID                                DISK USAGE   CONTENT SIZE   EXTRA
docker/getting-started:latest        d79336f4812b                     73.9MB       21.4MB         U
hello-world:latest                  d4aaab6242e0                     25.9kB       9.52kB         U

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker history docker/getting-started:latest
IMAGE      CREATED      CREATED BY                                      SIZE      COMMENT
d79336f4812b 3 years ago COPY /app/site /usr/share/nginx/html # build... 4.78MB    buildkit.dockerfile.v0
<missing>    3 years ago COPY /app.zip /usr/share/nginx/html/assets/a... 1.78MB    buildkit.dockerfile.v0
<missing>    3 years ago /bin/sh -c set -x && apkArch="$(cat /etc... 31.5MB
<missing>    3 years ago /bin/sh -c #(nop) ENV NJS_VERSION=0.7.9 0B
<missing>    3 years ago /bin/sh -c #(nop) CMD ["nginx" "-g" "daemon... 0B
<missing>    3 years ago /bin/sh -c #(nop) STOP_SIGNAL SIGQUIT 0B
<missing>    3 years ago /bin/sh -c #(nop) EXPOSE 80 0B
<missing>    3 years ago /bin/sh -c #(nop) ENTRYPOINT ["/docker-entr... 0B
<missing>    3 years ago /bin/sh -c #(nop) COPY file:e57eef017a41ca7... 16.4kB
<missing>    3 years ago /bin/sh -c #(nop) COPY file:abcbcf84dc17ee44... 12.3kB
<missing>    3 years ago /bin/sh -c #(nop) COPY file:5c18272734349488... 12.3kB
<missing>    3 years ago /bin/sh -c #(nop) COPY file:7b307b62e82255f0... 8.19kB
<missing>    3 years ago /bin/sh -c set -x && addgroup -g 101 -S ... 6.73MB
<missing>    3 years ago /bin/sh -c #(nop) ENV PKG_RELEASE=1 0B
<missing>    3 years ago /bin/sh -c #(nop) ENV NGINX_VERSION=1.23.3 0B
<missing>    3 years ago /bin/sh -c #(nop) LABEL maintainer=NGINX Do... 0B
<missing>    3 years ago /bin/sh -c #(nop) CMD ["/bin/sh"] 0B
<missing>    3 years ago /bin/sh -c #(nop) ADD file:S87cae71969871d3c... 7.73MB
```

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker system df
TYPE                TOTAL        ACTIVE        SIZE        RECLAIMABLE
Images              2            2            74.22MB     74.22MB (100%)
Containers          5            1            172kB       90.11kB (52%)
Local Volumes       0            0             0B          0B
Build Cache         0            0             0B          0B
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker exec -it 82aeb1f9729d sh
/ # exit
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker stop $(docker ps -q)
82aeb1f9729d
```

Создание контейнера docker/getting-started:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~$ docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
1e35f6679fab: Pull complete
cb9626c74200: Pull complete
9b6f639ec6ea: Pull complete
c158987b0551: Pull complete
ee68d3549ec8: Pull complete
4f7e34c2de10: Pull complete
b6334b6ace34: Pull complete
f1d1c9928c82: Pull complete
33e0cbbb4673: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
291f0b9314f0093e5c9b2f14cab8df5ca303706e0a55eba9b68528f513bbe21d
```

Открытие контейнера в браузере и изучение tutorial:

← → ↻ http://localhost/tutorial/ 🔍 Search 🐈 docker/getting-started 3.0k Stars

Getting Started

Table of contents

- The command you just ran
- The Docker Dashboard
- What is a container
- What is a container

The command you just ran

Congratulations! You have started the container for this tutorial! Let's first explain the command that you just ran. In case you forgot, here's the command:

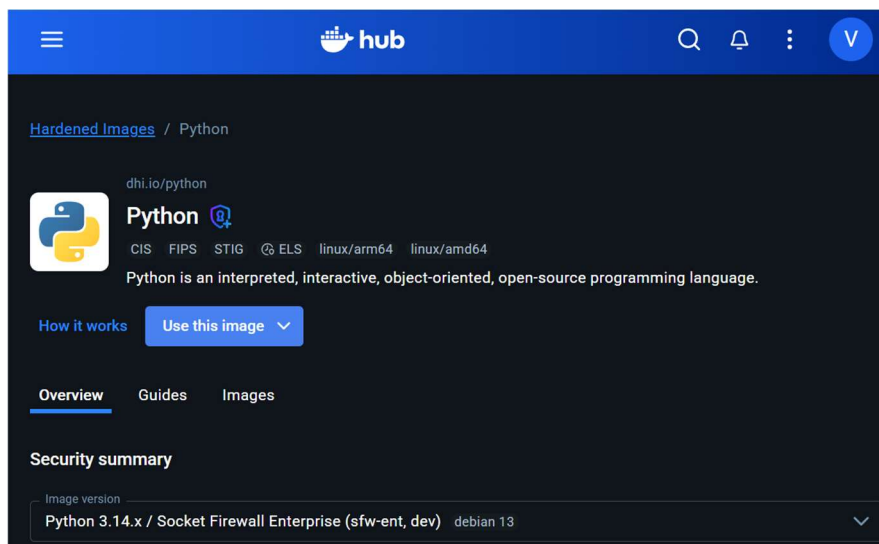
```
docker run -d -p 80:80 docker/getting-started
```

You'll notice a few flags being used. Here's some more info on them:

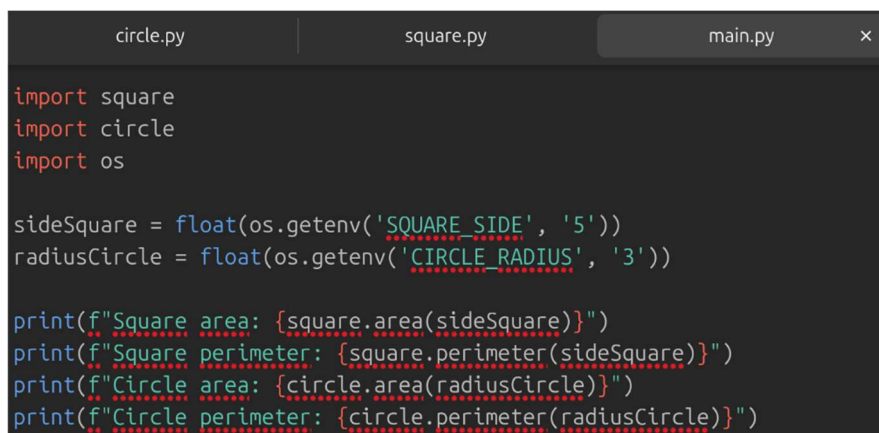
- `-d` - run the container in detached mode (in the background)
- `-p 80:80` - map port 80 of the host to port 80 in the container
- `docker/getting-started` - the image to use

3. Создание docker image, который запускает скрипт с использованием функций из https://github.com/smartiqaorg/geometric_lib:

3. а. Регистрация на DockerHub и выбор необходимых для проекта образов:



Реализация скрипта на основе репозитория:



3. б. Создание Dockerfile для реализации сборки Docker образов:



3. с. Создание и тестирование Docker-контейнера:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/geometric_docker/project$ docker build -t geometric_app:1.0 .
[+] Building 9.2s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 191B                               0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 2.9s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 694B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:c24e9effa2821a6885165d930d939fec2af0dcf819276138f11dd45e 4.6s
=> => resolve docker.io/library/python:3.11-slim@sha256:c24e9effa2821a6885165d930d939fec2af0dcf819276138f11dd45e 0.1s
=> => sha256:0b2bf04f68e9f306a8a83f57c6ced322a23968bf3d5acebc07e055c090240826 250B / 250B 0.3s
=> => sha256:3e731abb5c1dd05aef62585d392d31ad26089dc4c031730e5ab0225aef80b3f2 14.36MB / 14.36MB 1.6s
=> => sha256:5b09819094bb89d5b2416ff2fb03f68666a5372c358cfd22f2b62d7f6660d906 1.29MB / 1.29MB 1.0s
=> => sha256:119d43eec815e5f9a47da3a7d59454581b1e204b0c34db86f171b7ceb3336533 29.77MB / 29.77MB 3.2s
=> => extracting sha256:119d43eec815e5f9a47da3a7d59454581b1e204b0c34db86f171b7ceb3336533 0.5s
=> => extracting sha256:5b09819094bb89d5b2416ff2fb03f68666a5372c358cfd22f2b62d7f6660d906 0.1s
=> => extracting sha256:3e731abb5c1dd05aef62585d392d31ad26089dc4c031730e5ab0225aef80b3f2 0.3s
=> => extracting sha256:0b2bf04f68e9f306a8a83f57c6ced322a23968bf3d5acebc07e055c090240826 0.0s
=> [2/5] WORKDIR /app                                           0.1s
=> [3/5] COPY circle.py .                                        0.1s
=> [4/5] COPY square.py .                                        0.1s
=> [5/5] COPY main.py .                                          0.1s
=> exporting to image                                           0.8s
=> => exporting layers                                             0.5s
=> => exporting manifest sha256:3a23b4de4f0bd28a17794c380bdc4489dbb89c91132a062a81c8b71d9883c565 0.0s
=> => exporting config sha256:3ed8722e34c5bebeedaa3fd57904fb9d30eee1736976020b6c2381de21bbf6a5 0.0s
=> => exporting attestation manifest sha256:8d631cd1ac690b17e877e1cf5d7d21e996accd331d6ea963d09f7214eba52068 0.1s
=> => exporting manifest list sha256:f58f225e07a8b1c1211f8b17fddf7a5e9136dab6ca6ed52d0b77a6de3568a6c3 0.0s
=> => naming to docker.io/library/geometric_app:1.0              0.0s
=> => unpacking to docker.io/library/geometric_app:1.0          0.1s
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/geometric_docker/project$ docker run --rm geometric_app:1.0
Square area: 25.0
Square perimeter: 20.0
Circle area: 28.274333882308138
Circle perimeter: 18.84955592153876
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/geometric_docker/project$ docker run --rm -e SQUARE_SIDE=10 -e CIRCLE_RADIUS=
7 geometric_app:1.0
Square area: 100.0
Square perimeter: 40.0
Circle area: 153.93804002589985
Circle perimeter: 43.982297150257104
```

4. Создание многоконтейнерного веб-приложения со стек технологий в соответствии с индивидуальным заданием (вариант 9 – MongoDB, Express.js, AngularJS, Node.js):

Создание docker-compose.yml файла:

```
MEAN_APP
├── client
│   ├── .angular
│   ├── dist
│   ├── node_modules
│   └── src
│       ├── app
│       ├── assets
│       ├── favicon.ico
│       ├── index.html
│       ├── main.ts
│       ├── styles.css
│       ├── .angular-cli.json
│       ├── browserslistrc
│       ├── .gitignore
│       ├── angular.json
│       ├── Dockerfile
│       ├── nginx.conf
│       ├── package-lock.json
│       ├── package.json
│       ├── tsconfig.app.json
│       ├── tsconfig.json
│       ├── server
│       ├── .gitignore
│       ├── docker-compose.yml
│       └── package.json
└── docker-compose.yml

1  services:
2    mongodb:
3      image: mongo:7.0
4      container_name: mean_mongodb
5      ports:
6        - "27017:27017"
7      environment:
8        MONGO_INITDB_DATABASE: meandb
9      volumes:
10       - mongodb_data:/data/db
11     healthcheck:
12       test: echo 'db.runCommand("ping").ok' | mongosh localhost:27017/test --quiet
13       interval: 10s
14       timeout: 5s
15       retries: 5
16     networks:
17       - mean_network
18
19  mongo-express:
20    image: mongo-express:latest
21    container_name: mean_mongo_express
22    ports:
23      - "8081:8081"
24    environment:
25      ME_CONFIG_MONGODB_URL: mongodb://mongodb:27017
26      ME_CONFIG_MONGODB_ADMINUSERNAME: admin
27      ME_CONFIG_MONGODB_ADMINPASSWORD: pass
```

Сборка образов через команду docker compose build:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ cd ~/mean_app
docker compose build
[+] Building 2.5s (21/21) FINISHED
[+] build 2/2
✓ Image mean_app-backend Built 2.6s
✓ Image mean_app-frontend Built 2.6s
```

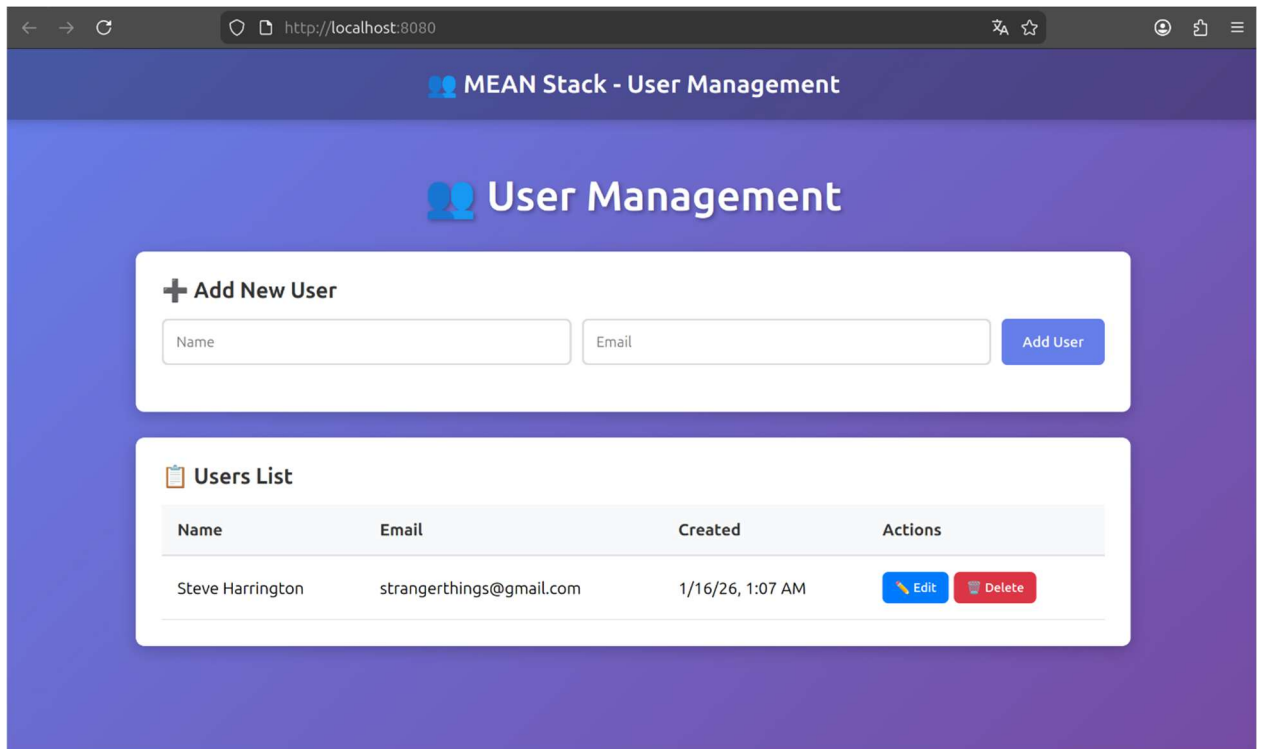
Запуск контейнеров через команду docker compose up -d:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker compose up -d
WARN[0000] No services to build
[+] up 4/4
✓ Container mean_mongodb Healthy 1.5s
✓ Container mean_backend Recreated 0.5s
✓ Container mean_mongo_express Running 0.0s
✓ Container mean_frontend Recreated 0.5s
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS
mean_backend mean_app-backend "docker-entrypoint.s..." backend 20 seconds ago Up 18 seconds (health: starting)
mean_frontend mean_app-frontend "/docker-entrypoint..." frontend 20 seconds ago Up 18 seconds
mean_mongo_express mongo-express:latest "/sbin/tini -- /dock..." mongo-express 13 minutes ago Up 13 minutes
mean_mongodb mongo:7.0 "docker-entrypoint.s..." mongod 13 minutes ago Up 13 minutes (health: starting)
```

Просмотр логов backend- и frontend-контейнера через команду docker compose logs и отображение списка изображений:

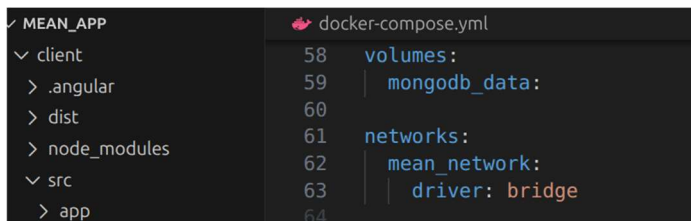
```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker compose logs backend --tail 30
mean_backend | > mean-backend@1.0.0 start
mean_backend | > node server.js
mean_backend | 🚀 Starting MEAN Backend...
mean_backend | MongoDB URI: mongodb://mongodb:27017/meanadb
mean_backend | (node:18) [MONGODB DRIVER] Warning: useNewUrlParser is a deprecated option: useNewUrlParser has no effect since Node.js Driver version 4.0.0 and will be removed in the next major version
mean_backend | (Use 'node --trace-warnings ...' to show where the warning was created)
mean_backend | (node:18) [MONGODB DRIVER] Warning: useUnifiedTopology is a deprecated option: useUnifiedTopology has no effect since Node.js Driver version 4.0.0 and will be removed in the next major version
mean_backend | 🚀 Server running on http://0.0.0.0:3000
mean_backend | ✅ MongoDB Connected
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker compose logs frontend --tail 20
mean_frontend | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
mean_frontend | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
mean_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
mean_frontend | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
mean_frontend | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
mean_frontend | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
mean_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
mean_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
mean_frontend | /docker-entrypoint.sh: Configuration complete; ready for start up
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker images | grep -E "mean|mongo|nginx"
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
mean_app-backend:latest a414b4833945 247MB 57.4MB U
mean_app-frontend:latest 4e01e362439d 92.7MB 26MB U
mongo-express:latest 1b23d7976f02 287MB 59.8MB U
mongo:7.0 8de574ce88c8 1.15GB 287MB U
nginx:alpine b0f7830b6bfa 93.1MB 26.8MB
```

Тестирование mean.app через браузер:

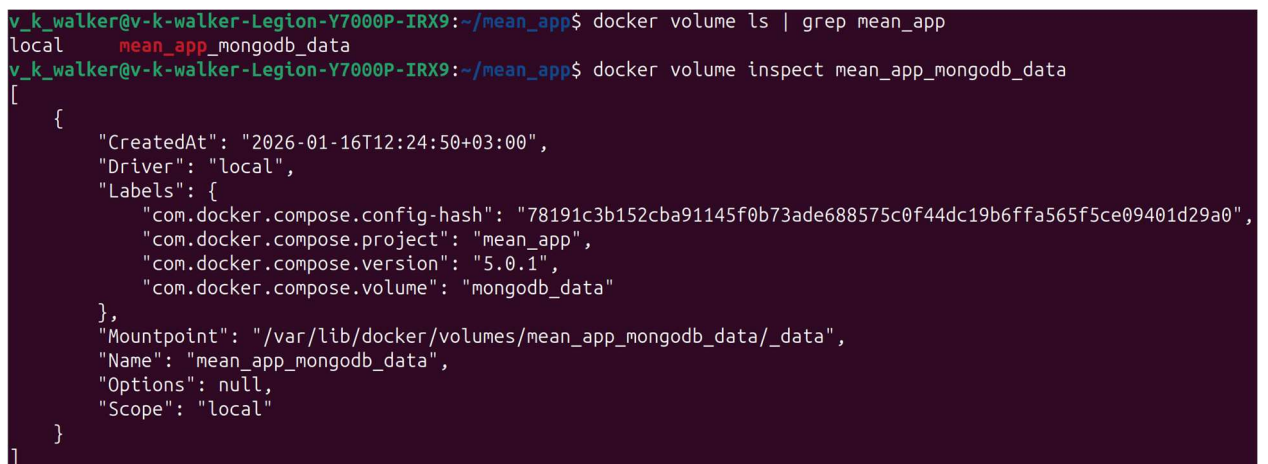


5. Настройка сети и тома для обеспечения связи между контейнерами и сохранения данных (исходные данные, логин, пароль и т.д.):

Указание в docker-compose.yml файле разделов volumes и networks:



Создание томов и сохранение данных:



Сети и связь между контейнерами:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls
NETWORK ID          NAME                DRIVER             SCOPE
c7b7234a48f5        bridge             bridge             local
e856e8847cd1        host               host               local
7ce49422860a        mean_app_mean_network bridge            local
fe32ca2779da        none              null               local
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect mean_app_mean_network
[
  {
    "Name": "mean_app_mean_network",
    "Id": "7ce49422860aef4cc35bbbc1d58a25bc202e11532f24932533f5feb234d299fb",
    "Created": "2026-01-16T12:33:39.490147378+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "IPRange": "",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {
      "com.docker.compose.config-hash": "32a13f298b03faa803a45375245ed8827446764ef399f516c0cd04e8611537ba",
      "com.docker.compose.network": "mean_network",
      "com.docker.compose.project": "mean_app",
      "com.docker.compose.version": "5.0.1"
    },
    "Containers": {
      "11491190bfe60c9cc337c023d304614bcbca1adb2f396b6434d97ed0a45ca5a3e": {
        "Name": "mean_frontend",
        "EndpointID": "0d4aa70eb66181a798f00099040c3fe67c8e29a7439d461923858551ad875488",
        "MacAddress": "82:15:9f:f2:af:16",
        "IPv4Address": "172.18.0.5/16",
        "IPv6Address": ""
      },
      "20fb94aa681907ee725c1d011dc4e4db302d6c84a941425ee335c6860618333": {
        "Name": "mean_backend",
        "EndpointID": "b1492dc60a04b6919653f149436bb50b8d0558494008e490cd8ece145d81f6e2",
        "MacAddress": "5a:8a:61:e0:fe:34",
        "IPv4Address": "172.18.0.4/16",
        "IPv6Address": ""
      },
      "ca2ab72ca6ede501139b8a9b449f14daaf4dbb6035e4a1c12325c72b508e77ef": {
        "Name": "mean_mongo_express",
        "EndpointID": "53fa160a2098956b0dfe77b37c27044c4acdc10bca164deacd2b6aa66bf1b716",
        "MacAddress": "1a:76:9a:13:09:91",
        "IPv4Address": "172.18.0.3/16",
        "IPv6Address": ""
      },
      "f120a4287ee650cc8d4d62341aa5903e6e8ca4d15e88d2730d6e1d07ba534f07": {
        "Name": "mean_mongodb",
        "EndpointID": "961b3ecfc25ef80b122138334c39ec84c43e7b5b30b4585f411c571fb5ad933c",
        "MacAddress": "7e:a6:cd:c5:4e:e7",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "172.18.0.0/16": {
              "IPsInUse": 7,
              "DynamicIPsAvailable": 65529
            }
          }
        ]
      }
    }
  }
]
```

Environment переменные (логины и пароли):

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ cat docker-compose.yml | grep -A 10 "^volumes:"
volumes:
  mongo_data:
    driver: local
  mongo_config:
    driver: local
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ cat docker-compose.yml | grep -A 5 "^networks:"
networks:
  mean-network:
    driver: bridge
    name: mean_app_network

volumes:
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$
cat docker-compose.yml | grep -B 2 -A 8 "environment:"
  container_name: mean_mongo
  restart: always
  environment:
    MONGO_INITDB_ROOT_USERNAME: admin
    MONGO_INITDB_ROOT_PASSWORD: secure_password_123
    MONGO_INITDB_DATABASE: meanapp
  ports:
    - "27017:27017"
  volumes:
    - mongo_data:/data/db
    - mongo_config:/data/configdb
--
  container_name: mean_mongo_express
  restart: always
  environment:
    ME_CONFIG_MONGODB_ADMINUSERNAME: admin
    ME_CONFIG_MONGODB_ADMINPASSWORD: secure_password_123
    ME_CONFIG_MONGODB_URL: mongodb://admin:secure_password_123@mongo:27017/
    ME_CONFIG_BASICAUTH_USERNAME: me_admin
    ME_CONFIG_BASICAUTH_PASSWORD: me_password_456
  ports:
    - "8081:8081"
  depends_on:
--
```

6. Размещение результата в созданный репозиторий в DockerHub:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker login
Authenticating with existing credentials... [Username: vkwalker31]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker tag mean_app-backend:latest vkwalker31/mean-backend:latest
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker tag mean_app-backend:latest vkwalker31/mean-backend:1.0
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker tag mean_app-frontend:latest vkwalker31/mean-frontend:latest
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker tag mean_app-frontend:latest vkwalker31/mean-frontend:1.0
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker images | grep vkwalker31
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
vkwalker31/mean-backend:1.0      cd798c12df34      247MB      57.4MB      U
vkwalker31/mean-backend:latest  cd798c12df34      247MB      57.4MB      U
vkwalker31/mean-frontend:1.0    53466622b3cb     92.7MB     26MB        U
vkwalker31/mean-frontend:latest 53466622b3cb     92.7MB     26MB        U
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker push vkwalker31/mean-backend:latest
The push refers to repository [docker.io/vkwalker31/mean-backend]
1074353eec0d: Mounted from library/nginx
063ba803406f: Pushed
c2b4197efb6c: Pushed
151e5bd84b3b: Pushed
c37f05a1102f: Pushed
3dcec9142507: Pushed
ca27fb533e38: Pushed
41b3afaea3b1: Pushed
de63f2020c4e: Pushed
latest: digest: sha256:cd798c12df34b70b2859ce0be0fef4c97d0954b862136dfacee23fd5eac2e size: 856
```

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker push vkwalker31/mean-backend:1.0
The push refers to repository [docker.io/vkwalker31/mean-backend]
3dcec9142507: Layer already exists
41b3afa3b1: Layer already exists
151e5bd84b3b: Already exists
de63f2020c4e: Layer already exists
ca27fb533e38: Layer already exists
063ba803406f: Layer already exists
c37f05a1102f: Layer already exists
1074353eec0d: Layer already exists
c2b4197efb6c: Layer already exists
1.0: digest: sha256:cd798c12df34b70b2859ce0be0fef9d4c97d0954b862136dfacee23fd5eac2e size: 856
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker push vkwalker31/mean-frontend:latest
The push refers to repository [docker.io/vkwalker31/mean-frontend]
0abf9e567266: Mounted from library/nginx
e096540205d5: Mounted from library/nginx
07d179254437: Pushed
1074353eec0d: Mounted from vkwalker31/mean-backend
25f453064fd3: Mounted from library/nginx
085c5e5aaa8e: Mounted from library/nginx
832e3f8a6025: Pushed
873b76cd8bd2: Pushed
da7c973d8b92: Mounted from library/nginx
33f95a0f3229: Mounted from library/nginx
567f84da6fbd: Mounted from library/nginx
latest: digest: sha256:53466622b3cb6b1afd67012b36b9a7792d34dec2182a023772e876d94683cf69 size: 856
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker push vkwalker31/mean-frontend:1.0
The push refers to repository [docker.io/vkwalker31/mean-frontend]
0abf9e567266: Layer already exists
25f453064fd3: Layer already exists
1074353eec0d: Layer already exists
da7c973d8b92: Layer already exists
e096540205d5: Layer already exists
07d179254437: Layer already exists
832e3f8a6025: Already exists
873b76cd8bd2: Layer already exists
33f95a0f3229: Layer already exists
567f84da6fbd: Layer already exists
085c5e5aaa8e: Layer already exists
1.0: digest: sha256:53466622b3cb6b1afd67012b36b9a7792d34dec2182a023772e876d94683cf69 size: 856
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker inspect vkwalker31/mean-backend:latest | jq '.[] | {Id, Repo'
{
  "Id": "sha256:cd798c12df34b70b2859ce0be0fef9d4c97d0954b862136dfacee23fd5eac2e",
  "RepoTags": [
    "mean_app-backend:latest",
    "vkwalker31/mean-backend:1.0",
    "vkwalker31/mean-backend:latest"
  ],
  "Created": "2026-01-16T01:05:54.697763152+03:00",
  "Size": 57381225
}
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker inspect vkwalker31/mean-frontend:latest | jq '.[] | {Id, Repo'
{
  "Id": "sha256:53466622b3cb6b1afd67012b36b9a7792d34dec2182a023772e876d94683cf69",
  "RepoTags": [
    "mean_app-frontend:latest",
    "vkwalker31/mean-frontend:1.0",
    "vkwalker31/mean-frontend:latest"
  ],
  "Created": "2026-01-16T01:05:48.028561868+03:00",
  "Size": 26019118
}
```

The screenshot shows the Docker Hub interface for the user 'vkwalker31'. The left sidebar contains navigation links: Repositories, Hardened Images, Collaborations, Settings, Billing, Usage, Pulls, and Storage. The main content area is titled 'Repositories' and shows a list of repositories within the 'vkwalker31' namespace. There are two repositories listed: 'vkwalker31/mean-frontend' and 'vkwalker31/mean-backend'. Both are public and have been pushed recently (5 and 6 minutes ago respectively). The interface also includes a search bar, a 'Create a repository' button, and a table with columns for Name, Last Pushed, Contains, Visibility, and Scout.

Name	Last Pushed	Contains	Visibility	Scout
vkwalker31/mean-frontend	5 minutes ago	IMAGE	Public	Inactive
vkwalker31/mean-backend	6 minutes ago	IMAGE	Public	Inactive

7. Изучение особенностей сетевого взаимодействия:

Получение информации о всех сетях, работающих на текущем хосте и подробности о каждом типе сети:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
c7b7234a48f5        bridge              bridge              local
e856e8847cd1        host                host                local
7ce49422860a        mean_app_mean_network bridge              local
fe32ca2779da        none                null                local
```

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "c7b7234a48f57cc11bab6e553f2d7b1859eb2c00a7475eb2aed88b24fa01cdf1",
    "Created": "2026-01-16T12:12:47.131898029+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "IPRange": "",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.17.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      }
    }
  }
]
```

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect host
[
  {
    "Name": "host",
    "Id": "e856e8847cd1278ad754e40e0ba2595fe4d3b03ea0f196c0e48ae5ac6b1bc679",
    "Created": "2026-01-13T19:47:57.16934433+03:00",
    "Scope": "local",
    "Driver": "host",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {}
    }
  }
]
```

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect none
[
  {
    "Name": "none",
    "Id": "fe32ca2779da58e6f7a84fd8d0b9a9cd6bef3708a037045aad985b1c05e5daf4",
    "Created": "2026-01-13T19:47:57.15503872+03:00",
    "Scope": "local",
    "Driver": "null",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {}
    }
  }
]
```

Создание и проверка custom-bridge сети:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network create --driver bridge my-bridge
9e89d327d325a9b0159f56813d9baf0ef224a74b0eed08791c8a28153a9f2e6c
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls | grep my-bridge
9e89d327d325      my-bridge        bridge          local
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect my-bridge
[
  {
    "Name": "my-bridge",
    "Id": "9e89d327d325a9b0159f56813d9baf0ef224a74b0eed08791c8a28153a9f2e6c",
    "Created": "2026-01-16T13:03:26.3879408+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "IPRange": "",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
```

Запуск Docker-контейнера в созданной сети и вывод информации о ней:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker run -d --name test-container-1 \
  --network my-bridge \
  --hostname test-host-1 \
  alpine sleep 1000
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
5c1f58ba4e0d: Download complete
644afed44dca: Download complete
Digest: sha256:865b95f46d98cf867a156fe4a135ad3fe50d2056aa3f25ed31662dff6da4eb62
Status: Downloaded newer image for alpine:latest
391ab4bba721b7ddf128585b02ab03454f09358bebc793e4aa0049c575dc0e23
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect my-bridge -f '{{range .Containers}}
Name: {{.Name}}
IP: {{.IPv4Address}}
MAC: {{.MacAddress}}
{{end}}'

Name: test-container-1
IP: 172.19.0.2/16
MAC: 96:3c:29:9e:bd:2f
```

Отключение сети от контейнера:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network disconnect my-bridge test-container-1
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect my-bridge
[
  {
    "Name": "my-bridge",
    "Id": "9e89d327d325a9b0159f56813d9baf0ef224a74b0eed08791c8a28153a9f2e6c",
    "Created": "2026-01-16T13:03:26.3879408+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "IPRange": "",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.19.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      }
    }
  }
]
```

Создание ещё одной сети bridge, вывод всей информации о ней и запуск в ней трёх контейнеров:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network create --driver bridge test-network
33de6d2234ce909f499435e4d382003b43369aa8c3385a23bd03a4542a305973
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker run -d --name web1 \
  --network test-network \
  --hostname web1 \
  alpine sleep 1000

docker run -d --name web2 \
  --network test-network \
  --hostname web2 \
  alpine sleep 1000

docker run -d --name web3 \
  --network test-network \
  --hostname web3 \
  alpine sleep 1000
d9c0e99560ed6331b9a66668ccf2c82ab5e1a0bca7895a33b1484ea711a97e13
77795e8111ca1f969d75d74516d1c629a670197ca9d7f6f4d43f8de09a099db6
e4e2047da442aff529e255fbfe05ecbc27af3ca39639ba78fc5148bf649045a1
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect test-network
[
  {
    "Name": "test-network",
    "Id": "33de6d2234ce909f499435e4d382003b43369aa8c3385a23bd03a4542a305973",
    "Created": "2026-01-16T13:09:47.838017255+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "IPRange": "",
          "Gateway": "172.20.0.1"
        }
      ]
    }
  }
]
```

```

    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "77795e8111ca1f969d75d74516d1c629a670197ca9d7f6f4d43f8de09a099db6": {
        "Name": "web2",
        "EndpointID": "6ec51ad5ab5daad47d3011846bbd2f612c06be7e60a5f26b4c4911726502017b",
        "MacAddress": "2a:3a:31:72:ae:73",
        "IPv4Address": "172.20.0.3/16",
        "IPv6Address": ""
      },
      "d9c0e99560ed6331b9a66668ccf2c82ab5e1a0bca7895a33b1484ea711a97e13": {
        "Name": "web1",
        "EndpointID": "fe9ccd531068c2a57795350576f4dd65ffaca0d77016f6eb465eb430d6d37a96",
        "MacAddress": "ae:60:d6:ed:f1:e9",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      },
      "e4e2047da42aff529e255fbfe05ecbc27af3ca39639ba78fc5148bf649045a1": {
        "Name": "web3",
        "EndpointID": "c13d93f205bcf86c8c6de839519768f26e3bb94030675d54189564ce26233894",
        "MacAddress": "12:5c:e0:ee:1f:82",
        "IPv4Address": "172.20.0.4/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.20.0.0/16": {
            "IPsInUse": 6,
            "DynamicIPsAvailable": 65530
          }
        }
      }
    }
  },
  "Status": {
    "IPAM": {
      "Subnets": {
        "172.20.0.0/16": {
          "IPsInUse": 6,
          "DynamicIPsAvailable": 65530
        }
      }
    }
  }
}

```

Подключение к первому контейнеру и пропинговка двух других из оболочки первого:

```

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect test-network -f '{{range .Containers}}{{printf "%-10s: %s\n" .Name .IPv4Address}}{{end}}'
web2      : 172.20.0.3/16
web1      : 172.20.0.2/16
web3      : 172.20.0.4/16

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker exec web1 ping -c 3 web2
PING web2 (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.065 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.099 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.101 ms

--- web2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.065/0.088/0.101 ms
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker exec web1 ping -c 3 web3
PING web3 (172.20.0.4): 56 data bytes
64 bytes from 172.20.0.4: seq=0 ttl=64 time=0.072 ms
64 bytes from 172.20.0.4: seq=1 ttl=64 time=0.118 ms
64 bytes from 172.20.0.4: seq=2 ttl=64 time=0.103 ms

--- web3 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.072/0.097/0.118 ms
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ WEB1_IP=$(docker network inspect test-network -f '{{range .Containers}}{{if eq .Name "web1"}}{{.IPv4Address}}{{end}}{{end}}' | cut -d/ -f1)
docker exec web2 ping -c 3 $WEB1_IP
PING 172.20.0.2 (172.20.0.2): 56 data bytes
64 bytes from 172.20.0.2: seq=0 ttl=64 time=0.092 ms
64 bytes from 172.20.0.2: seq=1 ttl=64 time=0.161 ms
64 bytes from 172.20.0.2: seq=2 ttl=64 time=0.196 ms

--- 172.20.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.092/0.149/0.196 ms

```

Создание собственной сети overlay, проверка на её создание, вывод о ней всей информации:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker swarm init
Swarm initialized: current node (9rqe4dgev7405rafnvdgkh2ih) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-1mi595qtx1a6yccxrnmujk6t4jqza4tziwhj6123zhd1b4xjvq-a4nrm52baz0xa1opbptvmgxyw 192.168.1.7:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network create --driver overlay overlay-net-1
ktg1e1qk2f7qtibfcufvorzly
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls | grep overlay
ujc96xhj8bit    ingress        overlay        swarm
ktg1e1qk2f7q    overlay-net-1  overlay        swarm
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect overlay-net-1
[
  {
    "Name": "overlay-net-1",
    "Id": "ktg1e1qk2f7qtibfcufvorzly",
    "Created": "2026-01-16T10:14:45.183886485Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "IPRange": "",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.driver.overlay.vxlanid_list": "4097"
    },
    "Labels": null,
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": [
          {
            "10.0.1.0/24": {
              "IPsInUse": 3,
              "DynamicIPsAvailable": 253
            }
          }
        ]
      }
    }
  }
]
```

Создание ещё одной сети overlay, проверка на её создание, вывод о ней всей информации и удаление:

```
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network create --driver overlay overlay-net-2
qq74gxv92hnwcybtrrci2igk1
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls --filter driver=overlay
NETWORK ID        NAME                DRIVER            SCOPE
ujc96xhj8bit      ingress             overlay           swarm
ktg1e1qk2f7q      overlay-net-1       overlay           swarm
qq74gxv92hnw      overlay-net-2       overlay           swarm
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect overlay-net-2
[
  {
    "Name": "overlay-net-2",
    "Id": "qq74gxv92hnwcybtrrci2igk1",
    "Created": "2026-01-16T10:17:36.700629587Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
```

```

        "Driver": "default",
        "Options": null,
        "Config": [
            {
                "Subnet": "10.0.2.0/24",
                "IPRange": "",
                "Gateway": "10.0.2.1"
            }
        ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
        "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
        "com.docker.network.driver.overlay.vxlanid_list": "4098"
    },
    "Labels": null,
    "Containers": {},
    "Status": {
        "IPAM": {
            "Subnets": {
                "10.0.2.0/24": {
                    "IPsInUse": 3,
                    "DynamicIPsAvailable": 253
                }
            }
        }
    }
}
]
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network rm overlay-net-2
overlay-net-2
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network ls | grep overlay
ujc96xhj8bit    ingress                overlay              swarm
ktg1e1qk2f7q    overlay-net-1         overlay              swarm

```

Попытка создания собственной сети host:

```

v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network inspect host
[
  {
    "Name": "host",
    "Id": "e856e8847cd1278ad754e40e0ba2595fe4d3b03ea0f196c0e48ae5ac6b1bc679",
    "Created": "2026-01-13T19:47:57.16934433+03:00",
    "Scope": "local",
    "Driver": "host",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {}
    }
  }
]
v_k_walker@v-k-walker-Legion-Y7000P-IRX9:~/mean_app$ docker network create --driver host my-host 2>&1
Error response from daemon: only one instance of "host" network is allowed

```